

Colorful Insights from an AI *Khipukamayug*

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Abstract

The color of each cord is often highlighted as one of the primary sites of meaning on an Inka *khipu* – the mostly undeciphered recording device of the Inka empire. However, while there are a variety of historical, ethnographic, and archaeological suggestions for how the semantics of color worked, none of these approaches systematically explains how Inka khipu cord colors denoted meaning. Instead, they present a range of fragmented, sometimes contradictory snapshots of a much larger, more complex system. This study aims to address the question: is there even a “system” of color that can be discovered and deciphered? If so, how did it work? Or is the sheer variety of uses discussed in prior studies an empirical indication that color had limited semiotic consistency for Inka *khipukamayugs* (“khipu makers/interpreters” in Quechua)? To answer these questions, I employ an AI khipukamayug – a custom “transformer” deep learning model trained to parse all of the cord colors in the Open Khipu Repository (OKR Team 2022) – to analyze the semantic space of Inka khipu cord color. My findings reveal a coherent system of color signs, wherein different colors occupy distinct, relationally-opposed semantic domains, with many colors demonstrating polysemy. This suggests a versatile set of color signs adaptable to various contexts and domains of representation, akin to other linguistic media. In the domain of khipus and beyond, this study points toward the importance of continued interaction with AI models as archaeologists learn (together with machines) to systematically represent and delineate the past.

Keywords— Inka, khipu, quipu, color, computational archaeology, word-sense disambiguation, deep learning

Cord Color and Meaning

We are told by Spanish chroniclers that the color of *kipu* cords played an essential role in representing information about the Inka social and cultural world. For instance, Garcilaso de la Vega reported that there was a one-to-one correspondence between cords colored yellow and the metal gold, those colored white and the metal silver, and finally red and warriors (1918 [1609]). Similarly, Antonio de la Calancha stated that *kipukamayus* (kipu makers or “animators” in Quechua) used the color black to represent time and that green stood for Inka troops who died in battle, among many other designations (1638). Generalizing their reports about how color signs worked, it may be said that colors primarily related to their object of meaning in either a one-to-one iconic or symbolic mode of representation (Peirce 1955). That is, a cord’s color referred to an object of meaning via resemblance (e.g. iconic use of yellow for the metal gold) or, in some cases, via cultural convention (e.g. black for time).

As of yet, however, these chronicler interpretations of color have not yielded a systematic framework for interpreting the colors of extant archaeological Inka kipu cords. Mackey notes, for instance, that a large percentage of all kipu cords are white. If Garcilaso was correct in his interpretation of white cords denoting silver, we would expect a wildly unrealistic amount of silver to have been counted by kipukamayus (1970, p. 57). Likewise, the vast majority of extant archaeological kipus seem to have been coded using only the natural colors of the yarn they were made from: the browns, tans, and whites of camelid and cotton fibers (Conklin 2002, p. 63). While the chronicler accounts provide us some sense of how we might interpret rarer colors like red, how do we parse the different shades of brown or the abundance of white cords? One explanation may be that iconism operated via a different modality than visual resemblance. Sabine Hyland (2017), for example, argues that cord color, in conjunction with cord fiber, phonetically denoted the Quechua names for cord colors in the post-conquest Collata kipu, which could then be used syllabically to record the names of indigenous social groups, called *ayllus*. However, this type of recording has not been demonstrated as a primary mode of representation outside of this restricted set of kipus. More than likely, if such colorful iconism and symbolism was also employed in Inka kipu color signs, it was used in conjunction with other modes of color signification.

For instance, Brokaw highlights how even an alphabetic script is based on both phonemic as well as non-phonemic conventions – e.g. numerals, punctuation, spaces between words (Brokaw 2010, p. 21). Accordingly, recent scholarship has revealed the importance of the “indexical” aspects of cord color – i.e. the way in which the contiguity of one cord color with another affects the meaning they convey (for more on the concept of indexicality and its relationship to icons and symbols, consult Peirce 1955, pp. 102–103). In post-conquest 20th-century kipu produced in Anchucaya, for example, Hyland (2016) found that kipu repeating bands of same-colored cords (“banded” kipu) recorded individual-level labor contribution data, while kipu with repeated sequences of different colors recorded aggregate, group-level labor data (“seriated” kipu). Furthermore, follow-up research has confirmed that these findings correlate more broadly with uses of the same adjacent color patterns in Inka labor accounting kipu (Clindaniel 2019). In the banded kipu at Anchucaya, each banded cord group corresponded to an individual – rank-ordered on the kipu by seniority – and the cords within the band to their individual contribution toward their group’s labor tasks (Hyland 2016, pp. 499, 501). Meanwhile, each color in a seriated cord group corresponded to a group’s contribution toward completing a particular communal la-

bor task (Hyland 2016, p. 505). This depiction suggests that color can refer both to individuals, as well as tasks. The distinct sense of a cord color was only determined in the context of the khipu (is it a banded or seriated khipu?) and with respect to cords that are contiguous to it (is it the same color or a different color? Is the cord in the same cord group or a different cord group?). While the symbolic and iconic aspects of these color signs no doubt had relevance for conveying meaning, the indexical relationships between cord colors appear to have also played a major role in interpretation. It seems probable that Inka banded and seriated khipus also worked in a similar way (even if the exact information they recorded in their banded and seriated groups of cords differed from that in the 20th-century Anchucaya context).

Yet, questions remain regarding the consistency and universality of color pattern meanings across different khipu genres. For instance, in 2013, Dr. Alejandro Chu and his team excavated 34 storehouse accounting khipus in a central storehouse facility at the South Coast Peruvian site of Inkawasi. Among these khipus were a set of khipus that “matched” one another (Urton and Chu 2015, 2019). Each pair of matching khipus in the set recorded calculations resulting in similar numerical values on each khipu. However, one khipu in a pair recorded the computation of the difference between a fixed numerical value (hypothesized to be a tax) and an initial value, while the other khipu recorded the computation of the difference between the resulting difference and initial value in the first khipu – resulting in the fixed numerical value. Noteworthy for this paper, however is that several of these storehouse accounting khipus (KH0503 and KH0491) seem to encode these operations in color seriated sequences of cords that behave in a different way than the aforementioned labor accounting khipus – suggesting genre-specific differences in the way color patterns denoted meaning.

In this storehouse accounting context, it has been argued that these color sequences may have instead denoted arithmetic operations (Clindaniel 2019). Cord color sequences in KH0503 mostly follow a consistent pattern: white denotes the minuend (the value from which another is subtracted), light brown the subtrahend (the value that is subtracted), while light brown mottled with medium brown denotes their resulting difference. However – it gets more complicated – in KH0491, light brown denotes the minuend, medium brown the subtrahend, and light brown mottled with medium brown continues to denote the resulting difference. Light brown – in a single archive, and in closely related khipus – seems to denote either that a value is the minuend or the subtrahend, depending on whether it is in a khipu where it is the lightest cord or not. When it is the lightest color on the khipu (KH0491), light brown takes on the role of minuend. However, when there is a lighter color on the khipu (white in KH0503), it instead denotes the subtrahend.

Recent work has questioned the exact set of arithmetic operations at play in these cord sequences – suggesting that white cords may be better interpreted as “sums” of the cords that fall in the same cord sequence of them. More generally, for instance, Medrano and Khosla 2024 have shown, across digitized khipus, that white cords often mark the start of a “sum cluster” – that is, a cord group in which each cord records the sum of cords elsewhere in the khipu (2024, pp. 12–13). This relationship explains less than half of the variation in color usage within sum clusters overall (41% of sum clusters begin with a white cord, compared to 32% of cord clusters overall), but it is at least a hint of one possible meaning (and indexical position) for white cords more generally. Nonetheless, even if we accept Medrano and Khosla’s suggestion that these white cords record sums, we must grapple with the fact that white seems to play the role of sum cord in the context of KH0503 and light brown seems to take on this same role in KH0491. That is, light brown seems to have been promoted from its role as a

summand cord in KH0503 to that of a sum cord in KH0491. At least in the limited context of these two khipus in the Inkawasi khipu archive, the meaning of a color seems to have relied on the overall “shade” of a cord group (and potentially khipu as a whole), determined by the lightest cord in a cord sequence. In particular, light brown did not solely rely on one-to-one iconic or symbolic characteristics of color, or even the same indexical relationships present in labor accounting “seriated” khipus. Instead, the sense of a color seems to have been partially informed by a shade-based indexicality with other neighboring cords.

More broadly, however, these findings raise questions about the role of indexicality in interpreting Inka khipu cord colors and the extent to which colors occupied distinct symbolic or iconic domains. If genre and contiguous cord patterns were so important for denoting meaning, was there room for iconism and symbolism of the sort described by the chroniclers and documented in post-conquest khipus? In an extreme case, for instance, we might posit that indexicality was the primary aspect by which colors were made meaningful. In this case, cord colors would not necessarily occupy universally distinct semantic domains; they would only be made meaningful by their distinctiveness from one another on a khipu. If this were the case, we might struggle to identify consistent enough uses of each color to delineate distinct semantic boundaries between them. Decipherment could only occur at small, local scales for particular khipus and archives.

Alternatively, a less extreme version of this theory might argue that colors occupied distinct iconic and symbolic domains, but had multiple, related senses that could be resolved via indexical relationships with other cords and colors. For instance, white cords might denote “silver” in a cord group context of yellow cords that denote “gold” and red cords that refer to “warriors,” as per Garcilaso, but refer to other (potentially related) concepts in the context of light and medium brown cords. In this case, khipu colors would behave something like “words” in other written and linguistic systems – with khipukamayus creatively employing the same signs to denote a variety of related concepts.

The present study aims to address these questions by leveraging recent computational advances in Natural Language Processing (NLP) and Deep Learning to train an AI model – an AI khipukamayus, if you will – that can identify and distinguish the meanings of cord colors, as well as evaluate their internal consistency. This approach draws parallels with challenges faced in NLP, where distinguishing between different word meanings is crucial for training machines to accurately understand, as well as generate speech and text. Specifically, we must first address the question: do cord colors have distinct meanings from one another? If we cannot delineate semantic boundaries between colors, this could be an indication that colors did not have consistent symbolic or iconic meanings – their meaning was mostly acquired via indexical relationships with other cord colors in a cord group and on a khipu. On the flip side, if it is possible to parse distinct semantic boundaries around each color, this might be indicative that they occupied distinct iconic and/or symbolic domains. Second, if we are able to identify distinct meanings for unique colors, how internally consistent are they? Did they have universal, one-to-one meanings (as suggested by the Spanish chroniclers)? Or did they have multiple senses, depending on the colors of the cords surrounding them, as the cases of color banding/seriation and light/dark arithmetic sequences at Inkawasi seem to suggest?

Word Sense Disambiguation in Natural Language Processing

In NLP, approaches that attempt to distinguish between- and within-word meanings are often generally framed as “Word Sense Disambiguation” algorithms. For instance, how do we instruct a computer to distinguish between the meaning of distinct words like “dog” and “cat” in a sentence? Additionally, while some words are monosemes – where each word has a unique meaning – natural language is full of polysemes – words with multiple associated senses – as well as homonyms – words that have multiple unrelated meanings (Vicente and Falkum 2017). Speakers of a language learn to distinguish between these words based on the context in which they are used, but how might a computer distinguish between the polysemic senses of the word “mouth” to describe a human body part, as opposed to its metaphorical sense in the context of the “mouth of the river?” Or, how might we help a computer delineate the semantic boundary between the differing uses of the homonym “bank” in the following two sentences?

I went down by the river *bank* to go fishing.

I went to the *bank* to deposit a check.

Computational approaches that rely on one-to-one dictionaries of terms-to-meanings (such as psychological inventories in Tausczik and Pennebaker 2010), or single representations of a term (e.g. Mikolov et al. 2013) collapse the many possible senses of a word into one quantitative representation and often make it difficult to correctly identify different usages of the same word or nuanced differences and similarities between words. To better account for these nuances, one successful strategy in NLP research has been to draw on the “distributional hypothesis” – assuming that words that are used in similar contexts tend to have the same meanings (Harris 1954). Likewise, as a corollary, we might also assume words that are used in consistently different ways can be interpreted to have distinct meanings (Miller and Charles 1991). These distinct meanings can then be disambiguated based on the context in which words appear – whether they be unique words, or the same word with multiple senses.

In this line of research, machine learning models are often used to learn different quantitative representations, or “contextual embeddings,” of words, based on the unique context in which each word is used in a text corpus. After such a model traverses all of the contexts in which each word in the corpus has been used and learns to effectively represent the words, clustering methods can be used to identify distinct clusters of meaning in the embedding space – an approach that has been shown to be highly effective for inducing the meaning of unknown words, disambiguating different senses of the same word, as well as parsing the semantic differences and similarities between different words (Garí Soler and Apidianaki 2021; Levine et al. 2020; Wiedemann et al. 2019). For instance, we might expect the embeddings for the word “bank” to appear in multiple distinct clusters – delineated based on the sense in which the homonym is used. Likewise, distinct words should also have their own unique clusters of contextual embeddings, with the distance between clusters in the embedding space defined by their relative difference in the semantic space of words. For instance, we might expect the classic structuralist oppositions like “raw” and “cooked” (*a la* Lévi-Strauss 1964) to be relatively far away from one another in the embedding space. Meanwhile, related words like “sushi” should be closer in the embedding space to “raw” than “cooked.”

In recent years, “transformer” deep learning models have proven themselves to be the state of the art in learning contextual embeddings via a series of “encoder” and “decoder” neural network layers (Vaswani et al. 2017). While the first transformers were designed with both “encoders” that learned embeddings from input sequences of data, and “decoders” that learned to generate sequences of data based on encoded embeddings and prompting text, in practice these components are often now decoupled. For instance, “encoder”-only models have been shown to be superior at performing interpretive AI tasks (as I will perform in this paper) because they are designed to encode contextual information about each token in a sequence – i.e. to generate contextual embeddings that can be used to disambiguate words and word senses. The Bidirectional Encoder Representations from Transformers (BERT) family of models is currently the most dominant set of encoder-based models (introduced in Devlin et al. 2019). On the flip-side, “decoder”-only models are particularly well-suited for generative AI tasks. For instance, Generative Pre-Trained (GPT) models (like ChatGPT) are decoder-based models that, given some prompt, will generate a response based on preceding text. For a more comprehensive introduction to encoder and decoder models and how they relate to other popular deep learning approaches, the reader is encouraged to consult Zhang et al. 2023 for an excellent introductory treatment.

Methods and Data

Disambiguating Meanings of Khipu Cord Colors

In the context of khipu cord colors, we can employ these methodological advances in word sense disambiguation to quantify the degree to which cord colors had distinct meanings from one another. If they did have distinct meanings from one another, we can then assess whether the colors had singular meanings (monosemes), or multiple distinct senses depending on the context of surrounding cords on a khipu or cord group (polysemes or homonyms). To perform this analysis, I trained a BERT model using the Python “Transformers” library (Wolf et al. 2020) to compute contextual embeddings for every recorded khipu pendant and top cord color in the Open Khipu Repository (OKR) – an open source, open access relational database that currently contains digital representations of 629 digitized Inka-style khipus from archaeological sites in the Andes, as well as museums around the world (OKR Team 2022). Note that all of the code necessary to replicate the model training and analysis is available as supplemental material (Clindaniel 2024).

The OKR is overseen by an advisory board consisting of Carrie Brezine, Iván Ghezzi, Sabine Hyland, Manuel Medrano, Jeffrey Splitstoser, as well as myself (I also currently administer the database). Khipus in the OKR come from provenances across the former Inka empire (from the North Coast of Peru down through the Northern Chilean coast), and feature a total of 6592 cord groups with colors recorded in them. Within these cord groups, there are 37,645 first-order (pendant and top) cords overall, and 48,681 individual color recordings (taking into account cords with multiple colors combined in them). Note, however, that the khipus that have been studied and digitized for inclusion in the OKR still make up a fraction of the 1,386 khipus that are known to exist in museums and private collections (as of 2021, when they were catalogued in Medrano 2021) and we continue to update the data in the repository as more is learned about khipus and additional khipus become available for study in collections and through active archaeological excavations. In addition to adding and

updating data, the advisory board promotes new initiatives aimed at maximizing data inclusivity and openness – such as renaming the khipus in the database from names reflecting investigator initials, such as “AS202” for the 202nd khipu studied by the Robert and Marcia Ascher, to a more consistent and neutral “KH0220” indicating the 220nd khipu indexed in the database (Brezine et al. 2024). In the present work, I use the “KH” names and numbers to refer to specific khipus. These “KH” numbers can be cross-referenced with their prior, corresponding investigator numbers in the OKR “khipu_main” table to facilitate comparison of findings with prior khipu studies.

The BERT training process involves a form of self-supervised learning, whereby the model learns from sequential input data directly, without access to manually-labeled data about the meaning of signs in these sequences. In the case of NLP, this is done by predicting missing words in a sentence (“Masked Language Modeling”). Whereas the base contextual unit of analysis in BERT language models is usually the sentence, in khipus, I presume that cord groupings have similar indigenous significance – i.e. they were grouped in a semantically meaningful fashion similar to a sentence and can also be productively used for mask-based modeling tasks.

To prepare the khipus in the OKR as input for the BERT model, I utilized Carrie Brezine’s Khipu Color Code System (KCCS) to transform cord colors into alphanumeric color codes (Urton and Brezine 2011). The KCCS provides groupings by hue and shade for every khipu cord color recorded in the OKR. For instance, hue “B” and shade “4” group together closely related “dark brown” colors that were recorded as eight different codes in earlier recordings by the Aschers (M. Ascher and R. Ascher 1978). These groupings aggregate slight variations in color into general categories that facilitate computational analyses across different khipus and archives. For instance, it is common for the “same” color to be recorded slightly differently depending on the cord material, observer perception, lighting conditions, as well as taphonomic processes.

I then separated each cord-specific code in a cord group by a blank space character and indicated cord group divisions with a special “[SEP]” indicator – as is conventional when training BERT models on text input with multiple sentences (see Figure 1). For cords with color combinations, I separated color codes with color combination symbols provided in the OKR. For instance, the symbols for the most common ways of combining multiple colors on a single cord – mottled (denoted in text by ‘:’), barber pole (denoted in text by ‘-’), and segmented (denoted in text by ‘/’) – were included between colors in color combination cords (with no spaces) so that the model could learn any potential differences between uses of colors in color combination cords as well as solid color cords. 15% of the cord colors in cord groups were randomly masked in preparation for the masked language modeling prediction task in training the BERT model. Note that Devlin et al. 2019 also performed a next sentence prediction task in training the original BERT model. In later BERT-family models, however, this task was found not to significantly improve (and in some cases, to even hurt) downstream task performance, and is often removed from the training process (Liu et al. 2019). For this reason, I exclusively used masked language modeling in my own khipu cord color training process.

The model then was trained to predict the masked cord colors based on the cords surrounding them – making predictions by learning to produce 768-dimensional vector representations of each cord color in context through a series of 12 encoder layers (each of which generates its own 768-dimensional vector that serves as input for the next encoder layer stacked atop it). I trained the model on 80% of the first-order (pendant and top) cord data in the OKR, iteratively improving performance by exposing it

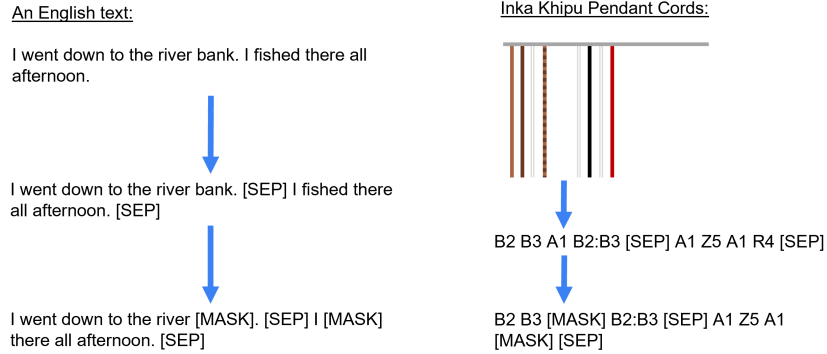


Figure 1: To train the BERT model on Inka khipu data, khipu cord groups were preprocessed in the same way as for standard masked language modeling tasks for contemporary writing systems. Cord groups were assumed to be meaningful groupings akin to sentences (adding “SEP” indicators between them) and cord color codes were randomly masked (with “MASK” indicators) in the same way as words.

to training data over and over again until prediction performance stabilized. I used the Optuna Python library to search over different hyperparameter configurations (for instance, the learning rate of the model) and identify the hyperparameters with the best performance across three random seeds (Akiba et al. 2019). Finally, I validated the model on the remaining 20% of the cord data, finding that the model performed similarly to its performance on the training data (Figure 2). Therefore, it appears that the model is not over-fitting to the training data and is relatively robust to unseen data. In other words, the model has learned to contextually represent cord colors with high enough precision that it can accurately predict masked cord colors in cord groups it has never seen before. Intuitively, this model can be thought of as an “AI khipukamayuk” that has learned to interpret (and produce) meaningful khipu signs based on extensive study.

To interpret the way in which the model has learned to represent each color, we must compute contextual embeddings for each cord, in the context of its cord group. Specifically, I followed Devlin et al. 2019 in concatenating the resulting vectors from the last four encoding layers of the model into a single contextual embedding of length 3072.

To map the semantic space of Inka khipu cord colors using these embeddings, we must answer two questions. Do different colors occupy distinct semantic domains in the embedding space? For instance, is light brown in a different location in the embedding space than medium brown? Once we have determined if there are differences between colors, we can then additionally assess whether these colors are likely to have had singular or multiple senses depending on the context in which they were used. For instance, are there multiple distinct areas of the embedding space that are home to white cords?

I employed a neural topic modeling strategy to perform between-color disambiguation, patterned off of BERTopic, as discussed in Grootendorst 2022: combining Universal Manifold Approximation and Projection (UMAP) to learn lower-dimensional (100-

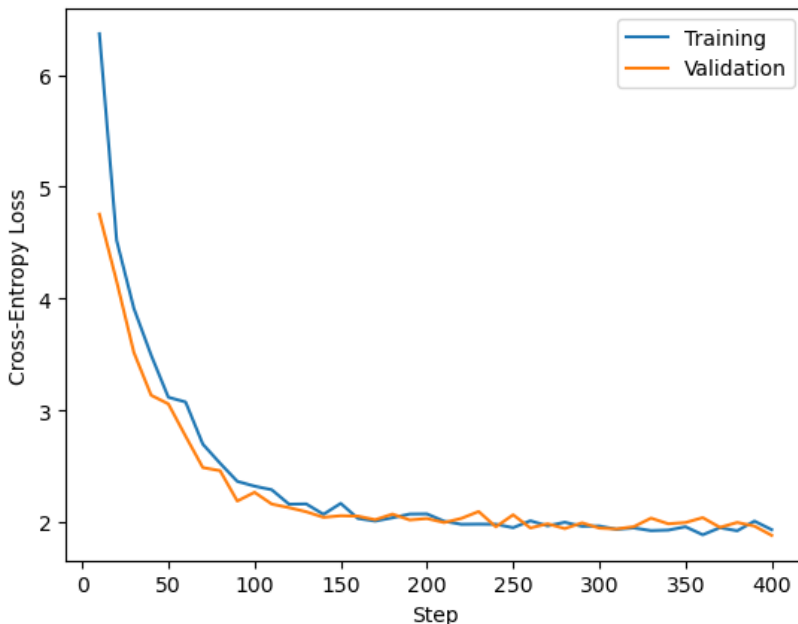


Figure 2: Training and Validation Cross-Entropy Loss (a representation of the number of prediction errors being made by the model as training progresses).

dimensional) representations of the 3072-dimensional contextual embeddings (McInnes et al. 2018), and then identify distinct clusters of cord color codes from these lower-dimensional representations using the HDBSCAN clustering method (McInnes et al. 2017). To conservatively identify large, coherent clusters within the corpus, I first identified clusters with a minimum cluster size of 250 (i.e. the smallest size grouping that can be considered a cluster is 250 closely related uses of a color). However, several color codes appear less than 250 times in the OKR – for instance medium yellow (Y3) only appears 132 times and medium gray (M3) appears even less at 93 times. Therefore, I additionally identified clusters with a minimum cluster size of 25 to study smaller sub-clusters inhabited by these less frequent color codes. Color codes that are clustered together may then be interpreted as meaningfully similar to one another, suggesting that the colors may have been conceived as the same (or semantically interchangeable) by Inka khipukamayus. In contrast, color codes that belong to their own exclusive (or near-exclusive) clusters might be interpreted as occupying their own semantic spaces and having their own distinct uses. Finally, I generated visualizations of the resulting clusters in two dimensions by applying UMAP once more to reduce the dimensionality of the embeddings.

With an understanding of between-color code distinctions and whether KCCS investigator codes are likely to map onto distinct semantic spaces for Inka khipukamayus, we can then proceed to focus on within-color comparisons. To assess the degree to which a given cord color was polysemic, we must answer question: how similar is a cord color embedding to itself in every context in which it appears?

To answer this question at a coarse-grained level, we can first count the number of distinct, aforementioned HDBSCAN clusters in which a color belongs as a primary member. If the same color is the primary member of several distinct clusters, they may be indicative that these are different senses of the same color – i.e. the color is polysemic, or perhaps a homonym. Likewise, if a color only appears in one cluster as a primary member, this may suggest that it is monosemic. It should be noted, however, that the number of distinct clusters is partially determined by the minimum cluster size used by the clustering algorithm (and, thus, the number of times a color appears in the OKR). Therefore, in order to compare the degree to which each color participated in polysemic behavior, I additionally used Python’s PyTorch library (Paszke et al. 2019) to collect all of the 3072-dimensional contextual embeddings for each cord color, compute the cosine similarity between each pair of embeddings (i.e. take the cosine of the angle between the two embeddings), and finally produce histograms to visualize the cosine similarity distributions for each cord color. Note that cosine similarity can roughly be equated to the concept of “correlation”: a cosine similarity of 0 between two embeddings means that they are highly unrelated (i.e. they belong to cord colors used in radically different contexts), whereas a cosine similarity of 1.0 means that the embeddings are highly related (cord colors used in identical contexts).

If a cord color had a singular, highly consistent meaning, we might expect to see its histogram of cosine similarities feature a single peak near 1.0 and have a high cosine similarity overall (i.e. all greater than 0.5). In this case, we might also expect the cord color to only fall in a single cluster, as per the HDBSCAN-identified clusters. In studies of contemporary written language, for instance, the number of contextual embedding clusters associated with a word has been shown to be correlated with the number of senses it has (Wilson and Marantz 2022).

On the other hand, if a color had multiple, distinct senses depending on the context of surrounding cords, we instead might expect to see multiple peaks in its histogram of cosine similarities, or a high density of values less than a cosine similarity of 0.5. Histograms with a left-skewed peak (i.e. a distribution with a long tail that extends below a cosine similarity of 0.5) might be interpreted as indicating that a minority of a color’s senses are divergent from the dominant sense. In both these cases, we might also expect the cord color to fall into multiple HDBSCAN-identified clusters (if the color appears enough times in the OKR to warrant the identification of unique clusters). While we will not know for sure whether a color with multiple senses was used as a homonym or a polyseme, we can make an educated guess based on its cosine similarity histogram. Studies of contemporary written language, for instance, have demonstrated that BERT models can correctly distinguish homonymic from polysemic uses of the same word in a way that matches human judgement – representing homonymic uses as more semantically distant than polysemic uses (Nair et al. 2020). Based on this prior work, we might then expect cord color homonyms to have cosine similarity histograms with peaks centered near zero (i.e. they should be right-skewed), whereas polysemous cord colors should have peaks centered around 0.5 and higher.

A potential concern in interpreting neural embeddings trained on colors recorded in an archaeological dataset is the potential for taphonomic transformation and subsequent mis-recordings of cords that once were dyed. For instance, Milillo et al. (2023) examined two Wari khipus at the Museum of World Culture in Gothenburg, Sweden and identified several cords that appear undyed in visible light, but were found to be dyed after closer inspection via multiband imaging, X-ray fluorescence (XRF), and High performance liquid chromatography-high resolution mass spectrometry (HPLC-HRMS). Likewise, another concern is that cords might be lost (leaving unknown la-

cunae in cord color sequences). For instance, in several Inka khipus in the British Museum, Rode et al. 2024 found that dark brown (B4) and black (Z5) cords were particularly susceptible to damage (and subsequent loss) – likely due to the use of an iron tannin dye, which can degrade the material to which they are applied.

At the aggregate level, however, these taphonomic concerns should not prohibit us from identifying between- or within-color distinctions. For instance, if we observe the above stated cosine similarity histogram patterns for even the darkest recorded colors in the KCCS (i.e. those cords that are not completely lost or dramatically faded), this should be considered strong evidence of monosemy and/or polysemy for at least a conservative sample of the cord colors that are used in Inka khipus.

For lighter-colored cords, some additional interpretive caution is required. A white cord (or any lighter-colored cord) that was originally a darker color (but faded) should have a contextual embedding that is distinct from those of other white cords if it is used in a distinct way from cords that were intended to be white. At the level of a cosine similarity histogram, this could result in an extreme left-skew for the histograms of lighter colors when there are a smaller relative number of misidentified, faded cords in comparison to the dominant use case of the lighter color. Alternatively, if there are a relatively large number of faded cords that are incorrectly recorded as a lighter color, we might expect there to be multiple distinct peaks in this histogram (i.e. evidence of many uses that contrast with those of the originally intended color). We can assess the degree to which these possibilities are cause for concern by consulting the between-color clusters identified by HDBSCAN, however. If lighter color embeddings systematically bleed into the semantic space of darker color cords (i.e. fall into the same clusters as darker colors), then this may suggest that they are faded and this is why they are being used in a way that is characteristic of a darker color. If lighter colors do not participate in the same clusters as darker colors, though, this would instead suggest that, at least at an aggregate level, the color recordings can be taken at face value.

Assessing the scale of contextually-specific meanings

Of the cords that have provenances recorded for their khipus, I then assessed the scale of contextually specific meanings within each color code. Are contextually-specific meanings specific to a particular khipu, or khipu archive? If this were the case, it may be evidence less of polysemy within a single system of color signs, than of idiosyncratic khipu signs specific to their creator (as per Quilter 2002). To answer this question, I identified individual khipus, as well as full archives that used a given color in ways that were highly divergent from a critical mass of other uses of the same color. Specifically, I computed average contextual embeddings for each color, aggregated at the cord group level. I then identified color-specific cord group embeddings that had less than 0.5 cosine similarity with other cord groups that employed that same color. Finally, I computed the overall percentage of color-specific cord groups that the cord group embeddings diverged from at the aggregate level of the individual khipu, as well as a complete archive. If the average divergence percentage of all cord groups on a khipu was greater than 25%, I called the use of a color in the khipu “highly divergent,” meaning that the khipu diverged greatly from the use of the same color elsewhere. Likewise, if the average divergence percentage of cord groups at a provenance-level was greater than 25%, I consider the use of the color in a particular provenance, or archive, to be highly divergent as well. If it is true that khipus were highly idiosyncratic devices, with meanings particular to their maker or archive, we might expect a large number of khipus and provenances to be recognized as highly divergent by these metrics.

To identify similar uses of cord colors in the context of a full cord group- or khipu-level sequence, I computed the average embeddings of cords at a cord group- and khipu-level, which can be interpreted as contextual embedding representations of cord color “phrases” or khipu-level “topics.” To identify related clusters of phrases and topics, I employed the same neural topic modeling strategy that I used to disambiguate cord color codes, but with smaller clustering thresholds to account for the smaller sample size inherent in aggregated cord groupings, as well as complete khipus (a minimum cluster size of 15 for phrases and 5 for topics). Phrases and topics that are clustered together can then be interpreted as meaningfully similar to one another. To identify the scale of their usage, I assessed the number of khipus and provenances overall that employed each phrase and topic. If the content denoted at the cord group- and khipu-level via color was specific to a particular khipu or provenance, this should be reflected by fewer khipus and archives employing a particular phrase or topic. On the flip side, if phrases and topics were widely used, they should appear in multiple khipus and archives.

Results

Unique Clusters of Meaning

The most frequently observed colors in the OKR all fall within unique clusters of meaning and 79% of all cords overall fall into one of 30 distinct clusters (see Figure 3). Note for instance, that the average cord color embeddings are in tightly-packed, distinct clusters for light gray (M2, 97% in cluster 12, 3% in cluster 13), black (Z5, 100% of black cords in cluster 4), medium blue (H3, 100% in cluster 3), medium green (G3, 99% in cluster 1, 1% not in a cluster) and dark green (G4, 92% in cluster 2, 8% not in a cluster).

All the shades of brown (B2-4) fall into their own semantic spaces as well. For example, 84% of light brown (B2) cords are clustered and the majority of the color is in cluster 17 (where its average is located). A minority of light brown cords (less than 15% overall) are in clusters 0 (below the average), 5 (to the right of the average), and 18 (to the upper right of the average). 77% of medium brown (B3) cords lie in a cluster, with the majority – 39% of clustered cords – in cluster 21 (where its average is located as well), but a comparable share of clustered cords (34%) is also in cluster 20 (immediately to the left of cluster 21). The remaining medium brown cords that are clustered fall in the smaller clusters orbiting clusters 20 and 21 – clusters 6-8 (shades of dark orange in the plot), clusters 14-15 (shades of yellow in the plot), and less than 1% in clusters 2-3. 100% of dark brown (B4) cords are clustered, with the majority – 87% – centered around the average (cluster 11), a smaller number – less than 13% – split across clusters 9-10 (the shades of light orange around cluster 11), and only 5 cords (0.1%) in cluster 4.

It might be contested that the multiple clusters associated with each shade of brown are simply an artifact of the aggregated KCCS codes (B2 accounts for 5 gradations of light brown in the Ascher color coding system, for instance; B3 corresponds to 9 Ascher codes, and B4 to 8 codes). However, even if we disaggregate the KCCS codes into Ascher color codes, the more fine-grained codes still fall in the same clusters in roughly the same proportions. For instance, AB, YB, and RL (and all the other color codes aggregated under “B2” in the KCCS) are centered in cluster 17 (where the average light brown cord is located) – indicating highly similar usage overall. MB

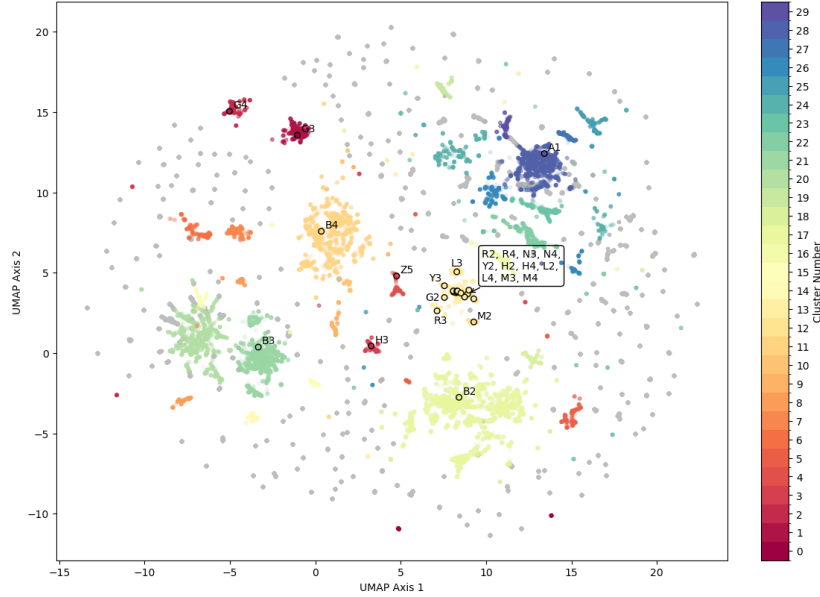


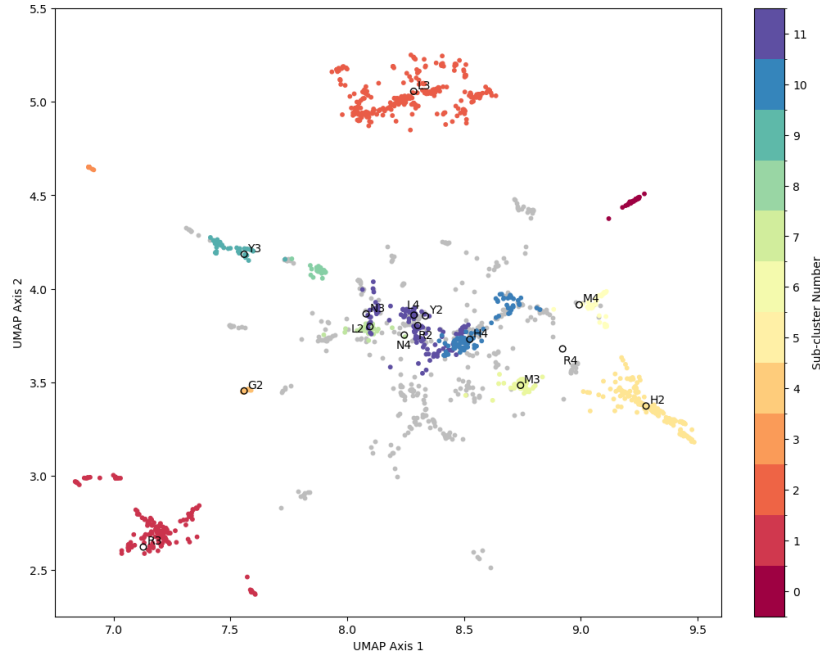
Figure 3: Cord color clusters (UMAP + HDBSCAN). Average embedding position for each color is indicated in the plot with a label. Outliers that were not assigned a cluster are presented in light gray.

and 0B (and all the other color codes aggregated under “B3”) are centered in clusters 20 and 21 – reflecting the overall proportions of medium brown cluster membership. All of Ascher codes aggregated as “B4” are likewise centered in cluster 11, as is the KCCS code as a whole. Therefore, there seems to be little evidence for systematic distinctions between these fine-grained color codes. Rather, regardless of how they are aggregated, these colors appear to be members of multiple different semantic clusters – providing initial evidence of polysemy.

The average white (A1) cord is in cluster 28 (and this is also its largest cluster – 34% of clustered white cords fall into this cluster). Note, however, that white is also the unique occupant of clusters 22-29 that surround this center (making up 89% of all clustered white cords in total). The remaining 11% falls in clusters 16 and 19 (slightly less than 11%) and cluster 0 (0.4%). Note, however, that only 66% of white cords in the OKR fall in clusters, which may indicate a variety of atypical uses that are not commensurable to how other colors are used. None of these uses overlap with darker colors, though, so there is limited evidence that this is a result of taphonomy or investigator perception alone. It is more likely that that white was used in multiple, related senses – i.e. it was polysemic as well.

All of the remaining colors have more than 95% of their cords as members of a cluster and nearly all of these cord colors are in cluster 13. This is a less frequently visited area of the semantic space (only 1776 cord embeddings are clustered in cluster 13, compared to cluster 21, for instance, where over 10,000 B3 cords are members) – perhaps indicating that these colors are part of a shared semantic field with more specialized meanings. Accordingly, all of the colors in this cluster appear infrequently

in the OKR overall. With the exception of H4 (dark blue; which occurs 92 times in sub-cluster 10), the other colors occur less than 50 times. If we permit HDBSCAN to identify smaller minimum cluster sizes (of 25 related cords, as opposed to 250), however, we can better disambiguate these lesser-used cord color sub-clusters within cluster 13 (see Figure 4). Most of the colors separate into their own sub-clusters, although there is a great deal of overlap in sub-clusters 10-11 – perhaps indicating related (non-disambiguable) uses for these colors. There are hundreds of sub-clusters overall (488 across the full embedding space), making it impractical to visualize all of them here. However, detailed printouts of sub-cluster membership for cluster 13, as well as all of the clusters displayed in Figure 3 are available in the supplemental materials.



– e.g. the colors in cluster 0 and cluster 3 – or do not cleanly fall into a sub-cluster – e.g. medium brown in cluster 2. In all of these cases, this may indicate a limited semantic fluidity whereby colors in neighboring clusters can be perceived to have similar meanings. These overlaps are limited in scope, however, and do not seem to indicate systematic taphonomic influence or misrecordings – if anything, the colors seem to be used in consistent ways overall and form relatively rigid semantic boundaries.

Semantic Relationships Between Colors

Figure 3 displays a highly reduced, 2-dimensional point representation of color embeddings, which, while valuable for visualizing high density clusters, is not necessarily a reliable aid for measuring distance between any two complete 3072-dimensional embeddings. By taking the cosine similarity between each pair of average color embeddings, though, we can more precisely assess how similar colors are to one another, as well as which colors are most semantically distant. In Figure 5, we can see that white (A1), light and medium brown (B2-3), as well as black (Z5) are most semantically distinct from each other as well as all the other cord colors. Note that white and black seem to be particularly opposed to one another. Dark brown (B4), medium green (G3), dark green (G4), and medium blue (H3) are also distinct from other colors – albeit less so (in the 0.5 cosine similarity range with many of the colors in cluster 13). The similarity between dark brown to medium brown, to light brown, to white seems to decrease linearly – suggesting relative darkness is at least partially a key to understanding the gradations of distinctions between the shades of brown and white.

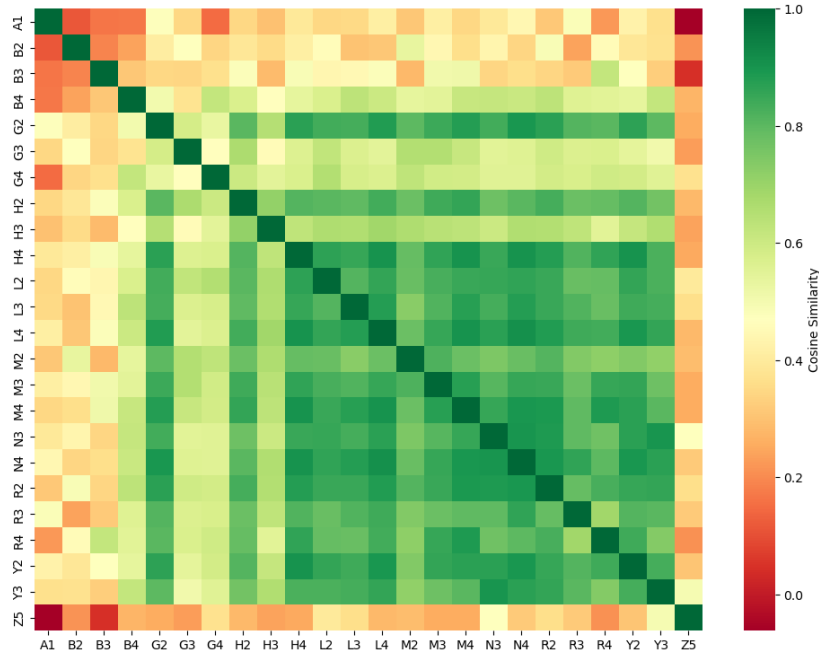


Figure 5: Cosine similarity between each pair of average color embeddings.

Light gray (M2) is similar to the colors in cluster 13, but less so than other colors in the cluster (reflecting that the majority of light gray cords are not members of cluster 13). The colors in cluster 13, are highly similar to one another, but also mostly set apart from the aforementioned colors. There are a few notable exceptions to these general rules, though. Note, for instance, that dark red (R4) seems to be used in similar ways to medium brown (B3), and light gray (M2) seems to be used in similar ways as B2. Meanwhile, white (A1) is quite unlike dark green (G4) and dark red (R4). Taking these contrasts in conjunction with white’s great distance from black (and gradations in meaning from B4-B3, B4-B2, B4-A1), this suggests possible contrasts in meaning by the sort of dark/light dichotomies discussed in (Clindaniel 2019).

Overall, Inka khipukamayusq seem to have primarily drawn from a “foundational set” of meaningful colors – white, light and medium brown, as well as black – that all had unique, relationally-opposed meanings from one another. This foundational set of colors makes up approximately 81% of cord colors in the OKR. To make additional distinctions from this foundational set of meanings, they used another “extension set” of colors – dark brown, medium green, dark green, medium blue, and light gray – all of which occupy their own clusters of meaning, but make up a more modest 15% of cord colors in the OKR. Finally, for additional distinction from the foundational and (to a lesser degree) extension sets of colors, they finally turned to the closely related colors in cluster 13, which we might call the “refinement set” of colors (making up 4% of cord colors in the OKR). These colors seem to only have been used sparingly in specialized contexts and may have been part of a shared semantic field given their close proximity to each other in the embedding space. Thus, in summary, not only do colors seem to have been semantically distinct from one another, but the distance between the colors suggests the way in which they related to one another – through a series of dark/light distinctions, as well as further distinctions between these foundational colors and more specialized extension and refinement sets of colors.

Within-color Sense-making

Given that there seem to be meaningful distinctions between the most frequently used KCCS color codes, as well as many of the less frequent color codes, I proceed under the assumption that these color codes provide at least a coarse-grained projection of the cord color semantic space. We can then drill down into each individual color code and interpret the degree to which the same color code was used in different ways – i.e. to what extent were Inka khipu cord colors polysemic?

Overall, the cosine similarity histograms for contextual embeddings of the same color suggest that only a minority of colors had singular, consistent use across cord group contexts. Note for instance, in Figure 6, cords colored dark olive (L4), light yellow (Y2), light red (R2), and dark orange (N4) were used in broadly consistent ways across all cord groups – indicating that they may have been monosemic. That being said, these colors are all in the center of cluster 13, sub-cluster 11, and appear infrequently overall in the OKR (dark olive appears 31 times, light yellow 10 times, light red 20 times, and dark orange 43 times). The small sample sizes may also explain how none of these colors have a peak exactly at 1.0 – even a few outlying embeddings will have a strong influence on the shape of these histograms. Nonetheless, in the restricted contexts in which these colors were used, they seem to have been employed consistently.

In contrast, the majority of colors seem to have had multiple senses. For instance, light green (G2) and dark red (R4) cords exhibited bimodal peaks (Figure 7), indicating

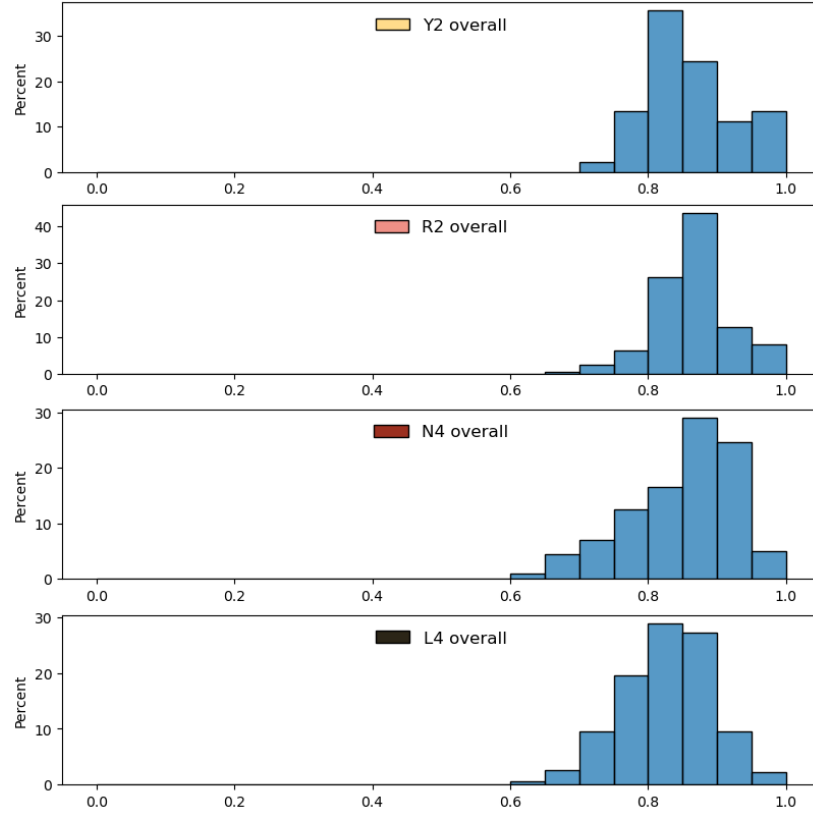


Figure 6: Cosine similarity histograms indicating consistent use across all contextual embeddings of a given cord color.

they may have had multiple, distinct (and distant) senses. These two colors also do not appear often in the recorded khipus in the OKR (light green appears 41 times, and dark red 35 times), but they are members of multiple sub-clusters in cluster 13. Light green, for instance, has its average embedding in sub-cluster 3, but also has membership in sub-cluster 11. Dark red appears in both sub-clusters 10 and 11 (although its average embedding does not fall in any identified cluster). Nonetheless, note that these bimodal peaks are centered around a cosine similarity of 0.5 and higher – a histogram pattern more consistent with polysemy (i.e. related senses) than homonymy.

The remaining colors all featured a single peak, with a long-tail pattern (and cosine similarity of less than 0.5 for a small selection of cords). This pattern suggests that they had a dominant sense (or set of closely related dominant senses), but also may have been used in more distant senses in a smaller number of cases. Again, given the left skew, however, these senses are likely to be polysemic in nature, as opposed to homonymic. There are two distinct types of long-tail distributions, however, that I argue have distinct interpretations – those that peak near a cosine similarity of 1.0 (i.e. 0.9-1.0) (Figure 8) and those that have a single peak less than 0.9 (Figure 9).

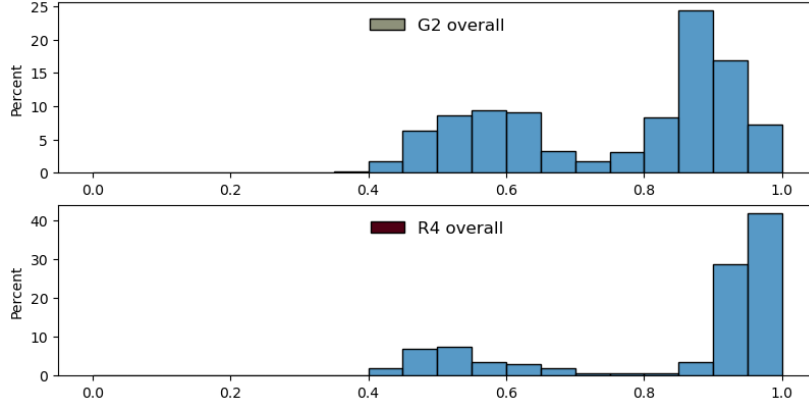


Figure 7: Cosine similarity histograms indicating bimodal, divergent uses across all contextual embeddings of a given cord color.

The long-tail distributions with a peak near 1.0 might be interpreted as colors with a dominant singular use (or set of closely related senses), along with a minority of other highly divergent use cases. White (A1), black (Z5), medium green (G3), and all of the shades of brown (B2, B3, B4) fit into this category (6 out of 18 colors with a long-tail distribution overall). For instance, light brown (B2), which was used in different, but related ways in the Inkawasi khipus KH0503 and KH0491 exhibits such a peak – indicating a dominant use, but also a minority of divergent uses. For reference, the overall cosine similarity of light brown in KH0503 versus KH0491 is 0.89 (i.e. a related usage, but not used in exactly the same way). Using light brown to refer to different, but related meanings outside of its core sense seems to have been less common across recorded khipus in the OKR, but there are a minority of cases that use the color in even more divergent ways than exhibited at Inkawasi. Likewise, it has already been observed in Figure 3 that white and all the shades of brown are centered in one cluster, but participate in multiple distinct clusters of meaning. Colors that belong to a single cluster, but have a dominant sub-cluster, alongside smaller sub-clusters also fall in this group. For instance, all black (Z5) cords fall into a single cluster (of which 68% are in a single sub-cluster and the remaining are spread across two additional sub-clusters) and the same is true for medium green (G3; 72% are in two related sub-clusters and the remaining spread across five additional sub-clusters).

Meanwhile, histograms with a long-tail and a peak less than 0.9 might be interpreted as colors where slightly different senses are common, although they are used in highly divergent ways in a minority of cases (see Figure 9). The majority of the cord colors fall in this category (12 out of 18 colors with a long-tail distribution). For instance, dark green (G4) appears in only one cluster, but has slightly less than 8% outside of this cluster (likely the reason for the long tail). Within the cluster, 54% of the dark green cords that are clustered fall in most dominant sub-cluster, with other sub-clusters each making up relatively equal percentages between 12-19% of the overall share of appearances. While light gray (M2) and medium blue (H3) are centered in their own clusters, their histograms also seem to indicate that their domains of meaning are internally more diffuse than the other foundational colors that are centered

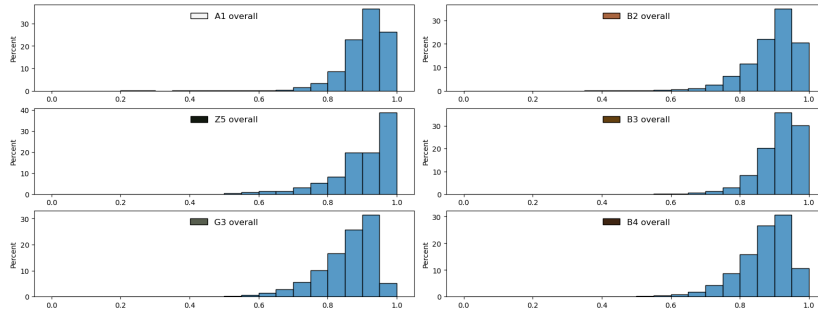


Figure 8: Cosine similarity histograms indicating a minority of divergent uses across all contextual embeddings of a given cord color, with a peak greater than 0.9.

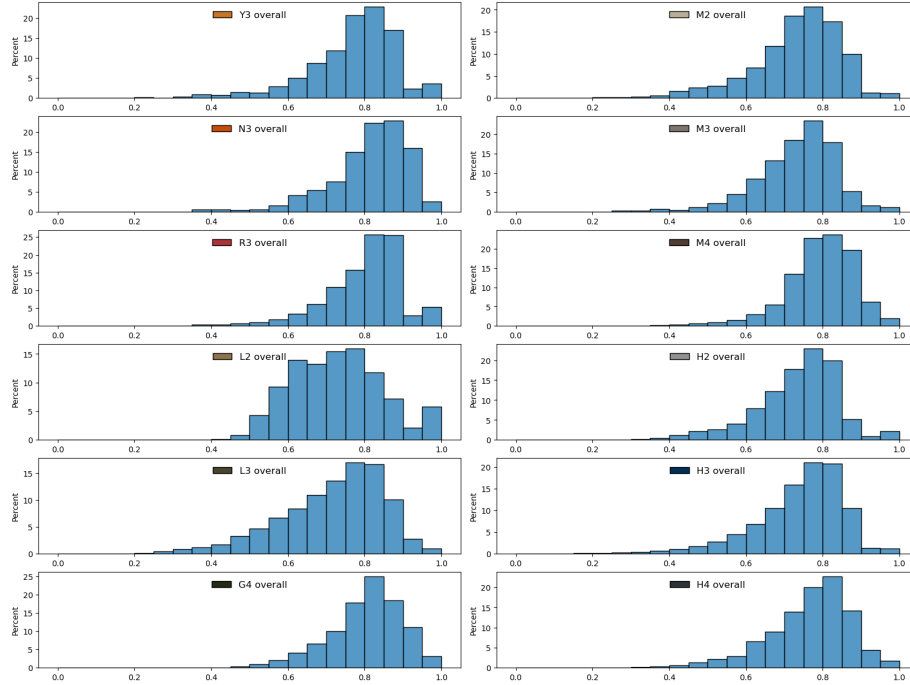


Figure 9: Cosine similarity histograms indicating a minority of divergent uses across all contextual embeddings of a given cord color, but with a peak less than 0.9.

in their own clusters. This is reflected to some degree in their cluster membership: light gray also appears to a lesser degree in another nearby cluster, and medium blue shares its cluster membership with 32 medium brown (B3) cords (which form a separate sub-cluster within its main cluster) – perhaps indicating more fluid membership

in the cluster as a whole.

There are similar distributions for all of the colors in cluster 13 – which should be expected given their small sample size in the OKR (i.e. the disproportionate effect a small number of outliers can have), as well as the fact that they all either belong to a multi-color sub-cluster or have occurrences across several sub-clusters within their dominant cluster. Take, for instance, light olive (L2), which has a histogram peak between cosine similarity of 0.6-0.8 and has membership in four different sub-clusters within (one of which is sub-cluster 11, that contains several of the other colors in the cluster). These colors all, thus, seem to have had slightly different senses depending on how they were mobilized.

Note that all of these patterns suggestive of polysemy (bimodal and long-tail) occur even in the darkest shade of colors, so these results do not seem to purely be a result of differential taphonomy, where a darker color with one meaning fades and is conflated with a lighter color (with another meaning) by the analyst. Likewise, both lighter and darker colors cluster in their own unique semantic spaces, so I find no evidence that either the between- or within-color variation is a result of taphonomic or observer bias alone.

Widespread Semantic Consistency

Importantly, each color code appears in multiple khipus and recorded provenances (Figure 10). Even highly divergent khipus or provenances only accounted for small percentages of the overall number of khipus that employed a color (Figure 11). The most divergent colors also tend to be the ones that appear in the fewest khipus and provenances overall. For instance, the color with the highest percentages in Figure 11 is dark red (R4), which is divergent in 1 out of 4 of the khipus that it appears in (KH0431, a khipu currently in the Ethnologisches Museum in Berlin, originally from Pachacamac). Note that this use of dark red is still in the expected cluster for the color (cluster 13), even though it does not fall neatly into an identified sub-cluster.

Medium yellow (Y3) closely follows dark red as one of the most divergent colors and is divergent in 1 out of 7 of the khipus (4 cords total) that it appears in (KH0348, a camelid khipu from Museo Temple Radicati with no recorded provenance). Interestingly, in conjunction with five divergent dark brown (B4) cords (in KH0289, from Museo Oro del Peru), these medium yellow cords are the only other colors to appear in cluster 4 alongside black (Z5). While taphonomy may be a concern for these particular cords (i.e. perhaps they are faded black cords), it is suggestive that both KH0348 and KH0289 are khipus with non-cotton cords – KH0348 is a camelid khipu and KH0289 is a human hair khipu. Could the material of a cord add semantic nuance that the color of a cord alone cannot capture? It could be that the same color on a different material allowed colors to participate in more nuanced semantic spaces that overlap with one another in the color semantic space (and may explain some of the unexplained variance in this “color-only” model).

The only provenances that stand out as systematically using a color in a different way from others are a single cord at Leymebamba (which is the only one out of seven provenances that use medium red – R3 – in the way that it does) and a single cord at Callengo (which is the only one out of seventeen provenances that use medium blue – H3 – in the way that it does). Nonetheless, these both represent only a small fraction of uses of these colors at other locations and only are divergent in their usage of these individual colors. Also, it should be noted that these are not divergent enough to be considered members of a different cluster than other uses of the same color. The use

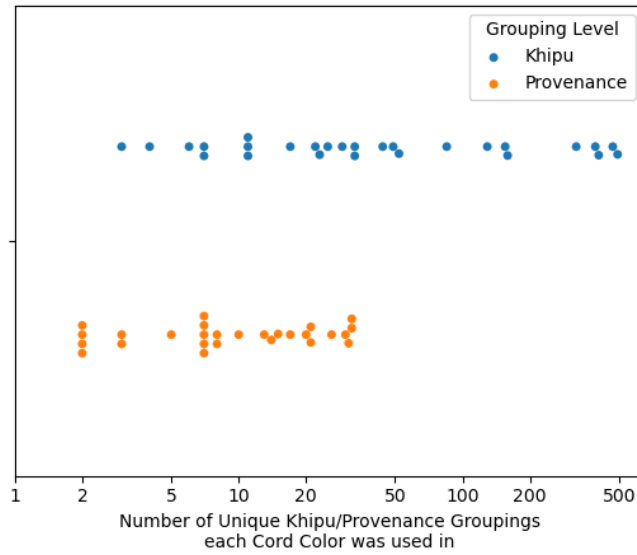


Figure 10: Swarm plot of the number of unique khipus or provenances (archives) a given cord color was used in. No cord color was used in fewer than two khipus or provenances.

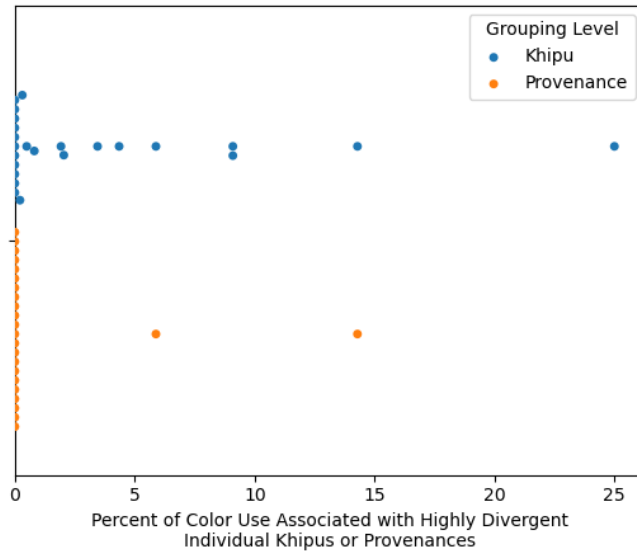


Figure 11: Swarm plot measuring the overall percentage of khipus or provenances (archives) that use a given color in a divergent way from other uses.

of medium blue (H3) at Callengo, for instance, is in both the expected cluster and sub-cluster for the color – albeit more distant than other cases. The use of medium red (R3) at Leymebamba is in the expected cluster, but different sub-cluster from all other instances of the color. The cord in question falls in cluster 13, sub-cluster 4 – which is otherwise the sole domain of light blue (H2). This indicates a slight difference in usage for the color, but still within the same general range of meaning.

Therefore, the observed semantic patterns are unlikely to purely be a result of idiosyncratic differences between individual khipukamayuqs or regional archival practices. Instead, these patterns seem to be broadly consistent at a more systematic, imperial level. It should be noted, however, that this does not rule out a restricted capacity for khipukamayuqs to include idiosyncracies in their khipus – either to record unusual meanings in a particular archive that were represented to a lesser degree in other regions (e.g. the use of medium red at Leymebamba), or to record meanings specific to a khipukamayuq (i.e. the unusual use of dark red in one khipu at Pachacamac).

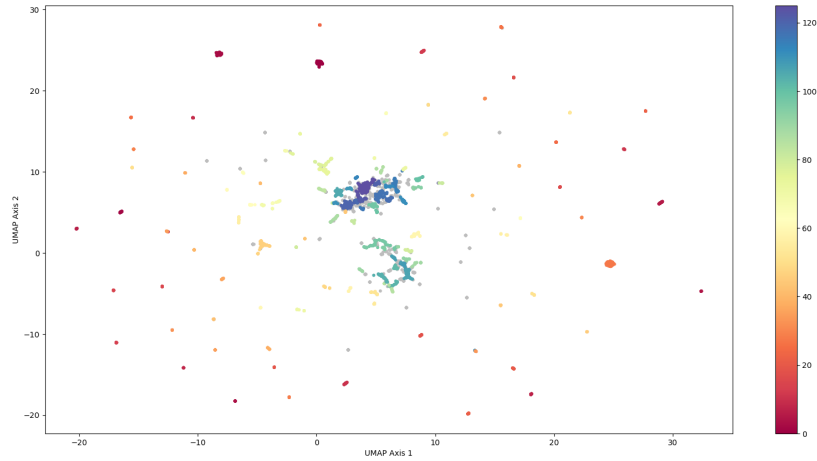


Figure 12: Cord Group “Phrase” Clusters identified by HDBSCAN and projected into two dimensions by UMAP. Outliers that were not assigned a cluster are presented in light gray.

Clustering aggregate cord groupings by cord color sequences (Figure 12) revealed that 87.2% of cord groups in the OKR belong to one of 127 distinct “phrase” clusters, with 95.2% of phrases appearing in more than one provenance and 84.8% of all cord groups with an identified phrase belong to a phrase that appears in four or more provenances. Likewise 88.0% of khipus in the OKR belong to one of 20 distinct “topic” clusters (Figure 13), with 100% of topics appearing in more than one provenance, and 95.1% of all khipus with an identified topic belonging to a topic that appears in four or more provenances. As one example of how the clustering algorithm has correctly distinguished different types of khipus, it has correctly distinguished the “light” and “dark” shades in the pair of Inkawasi khipus discussed prior sections (KH0503 is in topic cluster 6, KH0491 is in topic cluster 12), despite the use of similar colors overall. Thus, additionally, neither khipu-level cord color topics, nor cord group-level phrases seem to be specific to a particular archive. Instead, the way in which colors were

combined in cord sequences seems to have been broadly consistent across khipu, as well as archives.

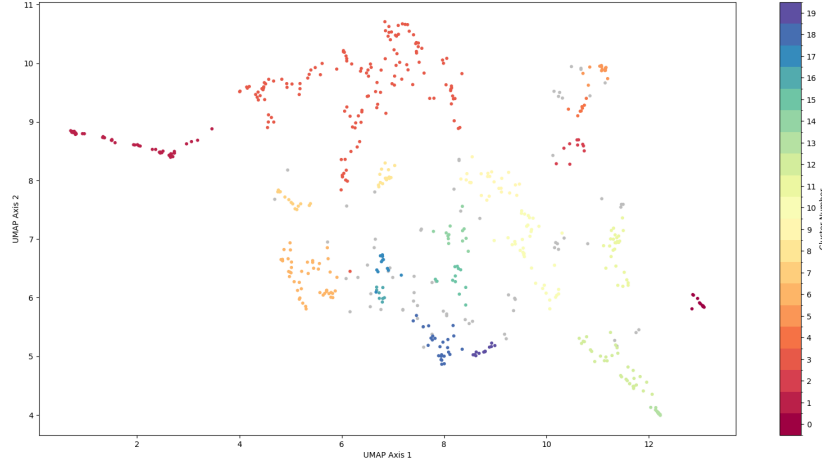


Figure 13: Khipu “Topic” Clusters identified by HDBSCAN and projected into two dimensions by UMAP. Outliers that were not assigned a cluster are presented in light gray.

Discussion

After exploring the semantic space of cord colors, our AI khipukamayuyq has discovered a distinct, “foundational set” of colors (white, light brown, medium brown, black) that fall within their own, relationally-opposed semantic domains. To form further semantic oppositions with this foundational set of colors, Inka khipukamayuyqs appear to have drawn from the distinct, but related semantic domains of an “extended set” of colors like dark brown, medium and dark green, medium blue, as well as light gray. Finally, for highly specialized semantic oppositions to the foundational set of colors, it seems that they drew from the remaining colors observed in Inka khipus to further refine and elaborate on the distinctions made within the foundational and extension sets of color – i.e. what I have termed a “refinement set” of colors which were all highly related to one another but generally distant from the foundational colors, as well as differentiated to a lesser degree from those colors in the extension set.

The extreme relational oppositions that are observed seem to mirror that which has been hypothesized as a theoretical model for Inka khipu color semiosis in Clindaniel 2019 – whereby colors denoted relationally opposed concepts that could be combined together in a variety of ways to refer to emergent, non-binary concepts that fall in between the two. This practice of relational opposition between colors has been demonstrated in a variety of domains in traditional Andean communities. For example, for some present-day khipukamayuyqs, the color white is associated with masculinity, and the color black is associated with femininity – distinguishing the two as complementary opposites (Arnold 2014, pp. 38–40). This has parallels in contemporary Bolivian weaving communities, where Cereceda notes that large bands (*chhuru*) of opposing

light and dark solid colors on textiles are often separated, or mediated, by two thin bands (*qallu*) of those same light and dark colors (1986, p. 168). According to weavers in the community, the thin, mediating bands form a category between the “masculine” light color and the “feminine” dark color—considered to be the “offspring” of the light and dark bands (1986, p. 169). In the present study, the diverging shades of black, brown, and white also seem to imply diverging meanings. For instance, the AI model has learned that dark brown is more semantically similar to medium brown than it is to light brown, than it is to white. Likewise, white is most dissimilar to black – its complementary opposite in the aforementioned studies.

Throughout Andean history, both cotton and llamas were specifically grown and bred to produce a variety of different natural shades of colors that were incorporated into textiles (Phipps 2020, pp. 33–34). These colors include, for instance, the various shades of brown, white, and gray, as well as black in camelids, and shades of green in cotton. Contrasts in fiber color had deep symbolic significance in Inka times. For instance, white llamas were said to have been sacrificed to the sun (which was connected to the Inka king) at the beginning of harvest (Cobo 1990 [1653]; Molina 1988 [1573]) and to have preceded the Inka ruler’s caravan (Sarmiento de Gamboa 1942 [1572], p. 40). In contrast, black llamas were said to have been starved and sacrificed during times of crisis (Murra 1978). Likewise different shades of “light” and “dark” colors seem to have held similar symbolic importance. For instance, Goepfert and Prieto found a Chimu burial with one light colored (beige) llama buried on top of a dark colored (brown) llama (Goepfert and Prieto 2016, pp. 204–206). They argue that these llamas were probably raised for ceremonial purposes as guides in the netherworld (207).

The results of this present study, however, suggest that the Inka *kipu* color semantic space was more nuanced than light/dark dichotomies alone. To produce more specialized meanings, for instance, *kipukamayuks* seem to have drawn from an “extension set” of colors, in which colors were used in ways that contrast with the “foundational set” of colors, but are not strictly opposed to one another on the basis of light/dark dichotomies. Such an extension set of colors would enable *kipukamayuks* to step outside a narrow range of complementary oppositions and denote a variety of additional concepts, distinct from what would be possible with the foundational set of colors. For even more specialized distinctions (and in even rarer occasions), *kipukamayuks* seem to have drawn from a single cluster of colors with highly related meanings – a “refinement” set of colors.

The use of these “extension” and “refinement” sets of colors brings to mind Antonio de la Calancha’s discussion of how colors may have been used to record the chronicles of the Inka kings – specifically, a campaign of the first Inka king, Manco Capac, in particular (1638, p. 91). In the system he described, *kipukamayuks* would draw from unique color-based semantic domains for concepts related to war (red), the Inka crown (crimson), Inka troops who died in battle (green), the spoils of war (where gold was denoted with yellow, and silver with white), and the sacred (blue). These colors could be combined together to form more complex ideas – e.g. indicating particular provinces, or even sacred festivals. For instance, Calancha wrote that a festival for the sun could be denoted on a *kipu* cord with a color combination of blue, with yellow and white – referencing the god who lives in the blue sky and creates gold (indicated with yellow) and silver (indicated with white). Importantly, however, in the absence of these distinctive colors, Calancha said that a *kipukamayuk* would interpret the cords in the chronicle *kipu* to denote that there was no king (e.g. in the absence of a crimson cord) or no sacred practices (e.g. in the absence of blue color

combination cords). At a high level, he seems to have described the same type of semiotic distinctions that our AI model has picked up on – whereby more specialized distinctions necessitated additional color associations beyond the typical, shade-based distinctions between whites, browns, and black in the foundational set of colors.

Notably, the refinement set contains nearly all of the colors that Calancha highlighted as being important to record a historical chronicle on a khipu: all the shades of red, several shades of green (G2 and L2-4), as well as multiple shades of blue (H2 and H4), oranges (N3-4), and yellows (Y2-Y3). We might ask the question then: is this the semantic region associated with concepts that denoted Inka royalty, the operation of the empire, and its religious festivals? Supporting this position, the refinement set of colors is the exclusive domain of all of the shades of red and yellow – colors that are often associated with Andean elites (e.g., the use of crimson to denote Inka royalty in Hyland 2020, p. 150) as well as deities (e.g., depictions of Paria Caca in Salomon and Urioste 1991, pp. 62, 68). Additionally, based on cosine similarity analysis, it appears that the colors in the refinement set were deployed in nearly identical contexts to one another – a pattern we would expect if these colors appear in contexts associated with Inka royalty, where a “bundle” of related concepts might be common (e.g. Inka royalty engages in wars, which produce casualties, and enable them to collect gold and silver). Likewise, if Calancha was correct in his assertion that the absence of these colors would denote the absence of their associated concepts, we would expect this closely related group of concepts to contrast strongly with the most commonly used cord colors on a khipu – as the refinement set contrasts with the foundational set in the embedding space. The rarity of these colors also seems to indicate highly specialized uses, which we might expect if they were only used to refer to concepts associated with the Inka crown and the sacred.

Note, however, that there are several distinct sub-clusters of oranges and yellows that may all have had their own unique (but related) senses. For instance, while the chroniclers typically associated yellow with gold (and thus with the spoils of war), it is possible it had additional senses more in line with the divine connotations of the color (or something else entirely that we do not yet understand). Likewise, given the many (11) semantic clusters that white is a member of, it is possible that white took on the sense of “silver” when it was used in contexts that involved refinement set colors, while maintaining several other, more common meanings in the context of colors in the foundational and extension sets. Thus, while these claims are still speculative and further research is needed to assess them more fully, the colors in the refinement set seem to align with expectations that they had meanings associated Inka royalty and the sacred – hinting that the descriptions of the Inka color system by Antonio de la Calancha may have been on the right track – if only in a highly restricted semantic domain.

Regardless of whether we understand the meanings of these color signs yet, however, it appears that Inka khipukamayucs were in widespread agreement about what each color sign meant and how it was used. Recall, for instance, that the contextual embeddings for each color, cord group, and khipu were generally consistent across different khipus and archives throughout the Inka empire. We might attribute some of this consistency to khipukamayuc training. The chronicler Martín de Murúa, for instance, wrote that there was a school in Cuzco where Inka elites and nobility from across the empire spent at least two years devoted to the interpretation and making of khipus (1987 [1590], pp. 376–377). Such a school would seem a prime location for future imperial administrators to learn to produce khipu color signs according to imperial conventions – enabling them to correctly interpret the unique meanings and

senses associated with each cord color. Additionally, archaeological evidence suggests that the Inka may have equipped *kipukamayuchs* with reference templates to ensure they followed the royal standard. Excavations in 2016 of a *kipu* production context at Inkawasi, for instance, identified colorful thread-wrapped sticks alongside other *kipu* production materials. Adjacent thread bands of color on the sticks account for the colors in over 75% of the color combination cords at Inkawasi and are believed to have been used as an aid in replicating color conventions at the site (Clindaniel 2019).

It is noteworthy that adjacent color combinations on the sticks at Inkawasi also strongly correlate with the opposing semantic domains discovered by the model in the present study. Nine out of the twelve unique color contrasts observed on the Inkawasi wrapped sticks (75%) also appear as distinct, opposing semantic domains in the present study. Specifically, there are contrasts between the foundational set of colors – white (A1) and light brown (B2; this pair appears on two separate sticks), white (A1) and medium brown (B3), light (B2) and medium brown (B3; this pair also appears on two separate sticks). Likewise, there are contrasts between the foundational and extended sets – white (A1) and dark brown (B4), light (B2) and dark brown (B4), as well as light brown (B2) and light grey (M2) – in addition to the foundational and refinement sets – light brown (B2) and light green (G2).

The remaining three unique color contrasts on the wrapped sticks are internal distinctions between shades of B2 (Ascher codes YB – light yellowish brown – and RL – light reddish brown, as well as YB and AB – light brown) and B3 (Ascher codes MB – moderate brown – and OB – olive brown). These adjacent color pairs seem to contrast with the findings from the OKR writ large, where there are no clear differences in cluster membership between Ascher color codes within the KCCS categories for B2 and B3. While it is plausible that there were archive-specific distinctions at Inkawasi, the distribution of cord colors at the site still reflect the same cluster membership overall as in the OKR – OB and MB are both centered in cluster 20 and AB, YB, and RL are centered in cluster 17. There are slight hints of distinctions between the Ascher color codes in the percentage of cords that fall within minority clusters (i.e. those clusters that feature only a minority of the instances of each cord color), but these should be taken with a grain of salt given the small sample size of these cord colors at the site. For instance, YB appears slightly more often in cluster 18 (8%) than RL (0%) and AB (1%). RL on the other hand, only appears in cluster 17. Likewise, OB has a higher percentage of its cords in cluster 8 at Inkawasi (24%) than MB (7%). However, again note the small sample sizes: YB only occurs 15 times in cluster 18 and OB only occurs 4 times in cluster 8. Nonetheless, the appearance of these color pairs on wrapped sticks warrants more detailed study of these colors (and their possible distinctions) in future research. It may be that the “same” color – B2 or B3 in the KCCS – had slightly different senses at Inkawasi based on fine-grained variations in cord color (or perhaps some other cord feature).

Overall, however, the strong correspondence with an Inka *kipu* cord production tool seems to provide strong supporting evidence that Inka understanding of how colors contrasted followed a similar logic to what the model has learned – enabling them to contrast meanings of solid colored cords, as well as form new emergent meanings through the combination of opposing semantic domains. This use seems to be in line with the use of these thread-wrapped devices in traditional Andean weaving circles. In such communities, weavers use wrapped sticks, called *musa waraña*, as semiotic tools that are used to standardize specific color combinations and sequences of warp threads in the process of warping up a loom (Arnold and Espejo 2012).

Using such a template for *kipu* colors would ensure the distinct semantic domains

of different colors remained consistent and interpretable across the empire, with some room for creativity in enabling different senses for the same color distinctions to take on new meanings for related concepts. The present study demonstrates, for instance, that within their semantic domains, most colors seem to have had multiple distinct senses depending on the context in which they were used – that is, they appear to be polysemic. Across the different colors, this polysemy produces at least 30 distinct senses across the colors that were used in Inka khipu cords. Recall, however, that using more granular clustering assumptions, the number of distinct senses could be in the hundreds (HDBSCAN discovered 488 sub-clusters). This high degree of polysemy has consequences for deciphering the Inka khipu color system – we need to take into account the context in which colors appear in order to resolve a meaning. At a higher level, however, it would also seem to provide empirical support for linking Inka khipus more closely to other forms of linguistic media – whereby speakers and writers of a language creatively extend existing words into related domains, rather than invent new words for each sense (Vicente and Falkum 2017). Studying these polysemic extensions in more detail may reveal a rich realm of senses in the signs of the Inka khipu – such as contronyms and metaphorical expressions – that gained their meaning via innovative correlation and contrast with existing meanings of colors and other sign vehicles.

Through this system of correlations and contrasts, further clustering approaches reveal that Inka khipukamayuks produced hundreds of unique “phrases” (127), which in turn formed 20 distinct “topics” that were used in khipus and archives across the Inka empire. As new khipus are discovered, one way of approaching their content may be to predict which clusters each khipu falls in and gradually resolve their meanings with association to other cord groups and khipus in diverse contexts. Further study of these more aggregated khipu- and cord group-level clusters may also reveal more information about how exactly colors took on different senses. For instance, was polysemy implicit and based entirely on context, or were there formal determinatives that helped interpreters narrow in on the sense of a particular cord color?

In previous work, it has been suggested for instance that khipus worked as “flow charts” (Hyland 2020) or nested, hierarchical sign systems (Clindaniel 2019) – whereby the meaning of signs like khipu-level color patterns, primary cords, and *kaytes* (the ornaments often affixed to the primary cord of a khipu) in turn informed the meaning of all the attached cords, which then informed the meaning of the knots on the cords, and so on. Was it possible to derive the sense of a pendant cord color based on such macro-level signs? Or were different senses indicated entirely at the cord-level? Perhaps there were no explicit indications on the khipu at all? Further computational research may, for instance, assess what the encoder model has learned to pay attention to in a sequence of colors when it encodes a color with a contextual embedding representation. This could then inform the relative performance of a model with knowledge of higher level information about a khipu (i.e. its primary cord and *kayte*). How much information do khipu-level signs actually provide in making sense of polysemic signs? Do khipu-level signs map onto the AI-discovered clusters of “phrases” or “topics” (perhaps indicating that all necessary information was provided by cord color patterns alone) or do primary cords and *kaytes* provide additional information that the current model has not learned?

Of course, it is also possible that such determinatives were not color-based at all, nor at higher hierarchical levels of the khipu. Recall, for instance, that the AI model placed several dark brown (B4) and medium yellow (Y3) cords into the same semantic domain as black (Z5). What may have distinguished these colors is not their color – but the material that was used to make the cord. One of these khipus

was made of human hair and another from camelid fiber – in contrast to the more commonly recorded cotton cords in the OKR. This use of cord material to denote meaning is not well understood in the Inka-style khipus of the OKR. However, in the 18th century Collata khipu, Sabine Hyland has argued that the incorporation of elements like silver and animal fiber into the khipu could be important indicators of meaning in conjunction with color (Hyland 2017, 2020). It seems possible that other cord- and knot-level modalities could also play a role in earlier khipus as well – perhaps acting as determinatives to resolve the sense of a particular cord color. Again, from a computational perspective, this becomes a question of the degree to which the incorporation of other modalities improves the performance of the model. Future research, for instance, might investigate the degree to which the incorporation of these other features adds new information to the model and helps to resolve the senses of cords that could not be cleanly clustered and delineated using cord color alone.

It should be noted that the OKR data includes excavated Inka khipus, as well as “Inka-style” khipus such as those from the Santa Valley that are likely to have been produced after the Spanish conquest – a chronology discussed by Ghezzi et al. in this volume. Of particular note in this study is that the Santa Valley khipus were not divergent in the way in which they used colors – they seem to be using the same color conventions as khipus across the OKR. This suggests that there may have been broad continuity in the use of certain khipu signs after conquest – highlighting the importance of continued, careful study of post-conquest khipus in relation to pre-conquest. Digitizing and incorporating khipus from different eras into the OKR is a priority of the OKR Advisory board (OKR Team 2022). This integration will enable additional computational comparisons between pre-Inka, Inka, and post-Inka khipus – potentially unlocking new insights into how colors and other sign vehicles were used, as well as how they fit into the broader tradition of khipu-making in the Andes. For instance, Splitstoser has hypothesized the potential for polysemic sign relationships in wrapped batons and cords at Cerrillos on the basis of color, as well as the properties of the yarn and cordage (Splitstoser 2014, p. 74). Might the polysemic senses of these “proto-khipus” map onto those discovered here for the Inka khipus? How about sequences of wrappings on Wari khipu cords? The colors of contemporary khipu cords?

Finally, for the field of archaeology more generally, this paper points to the methodological value of using self-supervised deep learning approaches to identify complex, hidden patterns in large archaeological datasets. The mask-based, neural modeling approach employed in this paper is widely used in the computational sciences for training AI models on large digital corpora of unlabeled sequence data without relying on expensive, time-consuming human labeling to perform large-scale content analyses. I would argue, however, that this approach to machine learning is particularly valuable for scholars working with digital archaeological corpora, which often consist of sequence data (images, text descriptions, stratigraphic top plans, etc.) with unknown labels. Self-supervised learning methods can help us parse large, unknown semantic spaces and build systematic theories for how meaning was constructed in the past.

Conclusion

In summary, this study has demonstrated the efficacy of employing an AI khipukamayuk model, based on a “transformer” architecture, to parse the semantic space of Inka khipu color signs. Synthesizing insights from ethnographic studies, archaeological

excavations, and chronicler accounts, my analysis reveals that khipu cord colors generally occupied distinct semantic domains with unique meanings. I have additionally identified three main sets of colors: a foundational set that recorded common, day-to-day concepts in terms of color shade-based relational oppositions, an extension set for broader meanings, and a refinement set for specialized contexts, such as royal and religious symbolism. The majority of these colors demonstrate polysemy, suggesting a versatile set of color signs adaptable to various contexts and domains of representation.

These findings underscore the importance of approaching khipus as a sophisticated writing system akin to other linguistic media, which AI models can learn to effectively represent. Moving forward, continued interaction with AI models holds promise for further synthesizing and deciphering the signs of the Inka khipu, as well as grappling with a variety of other complex archaeological research domains.

Data Availability Statement

All of the data necessary to replicate this analysis is available in the Open Khipu Repository, version 2.0.0 (OKR Team 2022). All replication code and output is available via Zenodo at <https://zenodo.org/doi/10.5281/zenodo.11218148> (Clindaniel 2024).

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